# IGCSE 0607 Practice Questions — E3.3

## Functions

**Total Questions:** 27 — **Total Marks:** 208 Marks

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**1** Functions $f$ and $g$ are defined as follows.

$f(x) = 2x - 5$, $g(x) = x^2 + 1$

**(a)** Find the value of $f(g(3))$.

..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**2** It is given that $f(x) = 3x - 2$ and $g(x) = \dfrac{10}{x}$.

**(a)** Find $f(4)$.

..... **[1]**

**(b)** Find $x$ when $g(x) = 2$.

$x =$ ..... **[1]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**3** The functions $f$ and $g$ are defined as $f(x) = 3x - 2$ and $g(x) = x^2 + 4$.

**(a)** Work out

**(i)**

hspace*0.8cm $f(-4)$

..... **[1]**

**(ii)**

hspace*0.8cm $gf(2)$

..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**4** The functions $f$ and $g$ are defined as $f(x) = 5x - 2$, $g(x) = x^2 + 3$.

**(a)** Find the value of $f(4)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $fg(2)$.

..... **[2]**

**5** Consider the functions $f(x) = 3x - 2$ and $g(x) = x^2 + 4$.

**(a)** Find $f(4)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $gf(2)$.

..... **[2]**

**6** Let $f(x) = 5 - 2x$, $g(x) = x^2 - 3$.

**(a)** Work out $f(-4)$.

..... **[1]**

**(b)** Work out $gf(2)$.

..... **[2]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**7** $f(x) = 4x - 3$, $g(x) = 2 - 5x$

**(a)** Work out $f(2)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Work out $gf(1)$.

..... **[2]**

**8** Let $f(x) = 4x - 3$ and $g(x) = 2x^2 + 1$.

**(a)** Work out $fg(3)$.

..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Solve the equation

$$f(x) = 17$$

$x =$ ..... **[2]**

**9** Two functions, $f$ and $g$, are defined as follows:

$$f(x) = 3x - 2 \qquad g(x) = x^2 + 1$$

**(a)** Work out $f(4)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[3]**

**10** The functions $f$ and $g$ are defined as $f(x) = \dfrac{3x+2}{x-1}$, $x \neq 1$, $g(x) = x - 2$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Solve the equation.

$f(x) = g(x)$

$x =$ ..... , $x =$ ..... **[4]**

**11** Consider the functions $h(x) = \dfrac{12}{x-1}$ and $k(x) = 2x + 3$.

**(a)** State the value of $x$ which must be excluded from the domain of $h(x)$.

$x =$ ..... **[1]**

**(b)** Find $k^{-1}(x)$.

$k^{-1}(x) =$ ..... **[2]**

**(c)** Find $hk(x)$, giving your answer in its simplest form.

$hk(x) =$ ..... **[2]**

**(d)** Solve the equation.

$h(x) = k(2)$

$x =$ ..... **[3]**

**12** Let $f(x) = \dfrac{4}{2x-3}$ for $x \neq 1.5$, and $g(x) = x + 5$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Solve the equation $f(x) = g(-3)$.

$x =$ ..... **[3]**

**(c)** Express $gf(x)$ as a single fraction in its simplest form.

$gf(x) =$ ..... **[3]**

**13** The functions $f$ and $g$ are defined as $f(x) = \dfrac{x+3}{x-1}$, where $x \neq 1$, and $g(x) = 2x - 5$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Show that $ff(x) = x$.

**[3]**

**(c)** Solve the equation $g(x^2) = f(3)$.

$x =$ ..... **[4]**

**14** $f(x) = \dfrac{3x+2}{x-1}$, $g(x) = \dfrac{2}{x+1}$

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Find $fg(x)$, giving your answer as a single fraction in its simplest form.

..... **[3]**

**(c)** Solve the equation.

$f(x) = f^{-1}(x)$

Give your answers in the form $a \pm \sqrt{b}$, where $a$ and $b$ are integers.

$x =$ ..... **[4]**

**15** The functions $f$ and $g$ are defined as follows:

$f(x) = \dfrac{3x+5}{x-1}, x \neq 1$, and $g(x) = x^2 + 1$

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

**(b)** Solve the equation $f(x) = x$.

$x =$ ..... , $x =$ ..... **[4]**

**(c)** Find $gf(2)$.

$gf(2) =$ ..... **[2]**

**16** The functions $h$ and $k$ are defined as $h(x) = \dfrac{2x+1}{3}$ and $k(x) = 5 - x$.

**(a)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

**(b)** Find an expression for $hk(x)$ in its simplest form.

$hk(x) =$ ..... **[2]**

**(c)** Solve the equation $h(x) = k(x)$.

$x =$ ..... **[3]**

**17** Functions $h$ and $k$ are defined as follows.

$h(x) = \dfrac{4}{x-3}$ for $x \neq 3$

$k(x) = 2x + 1$

**(a)** Find $k(h(5))$.

..... **[2]**

**(b)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

**(c)** Solve the equation $h(k(x)) = 2$.

$x =$ ..... **[3]**

**18** The functions $f$, $g$ and $h$ are defined as follows:
$f(x) = 2x + 3$, $g(x) = x^2 - 4$, $h(x) = 5^x$

**(a)** Find $h(-1)$ as a fraction in its simplest form.

..... **[1]**

**(b)** Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[3]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(d)** Find the value of $x$ when $h(x) = 125$.

$x =$ ..... **[1]**

**19** Two functions, $f$ and $g$, are defined as follows:

$$f(x) = \frac{5}{x-2} \qquad g(x) = x + 2$$

**(a)** State the value of $x$ that must be excluded from the domain of $f(x)$.

$x =$ ..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(c)** Solve the equation $f(x) = g(x)$.

$x =$ ..... , $x =$ ..... **[4]**

**20** $f(x) = 2x - 3$, $g(x) = x^2 - x - 7$

**(a)** Work out $g(-2)$.

..... **[2]**

**(b)** Find $gf(x)$ in its simplest form.

..... **[3]**

**(c)** Solve the equation.

$f(x) = g(x)$

$x =$ ..... , $x =$ ..... **[3]**

**21** Let $f(x) = \dfrac{2x+1}{3}$ and $g(x) = \dfrac{5}{x-2}$, where $x \neq 2$.

**(a)** Find $gf(x)$.
Give your answer as a single fraction in its simplest form.

..... **[3]**

**(b)** Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[3]**

**(c)** Solve the equation

$$f(x) = g(3)$$

$x =$ ..... **[3]**

**22** The functions $f$, $g$ and $h$ are defined as $f(x) = 4 - 3x$, $g(x) = \dfrac{5}{x-2}$, $x \neq 2$, $h(x) = x^2 - x - 7$.

**(a)** **(i)** Find the value of $f(0.5)$.

..... **[1]**

**(ii)** Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[2]**

**(b)** Find the range of $h(x)$.

..... **[2]**

**(c)** Solve the equation.

$h(x) = 2x - 1$ Give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

**23** Functions $f$, $g$ and $h$ are defined as follows.

$f(x) = 3x - 2$

$g(x) = x^2 - 4x$

$h(x) = \dfrac{2x+1}{x-1}$ for $x \neq 1$

**(a)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

**(b)** Find an expression for $g(f(x))$ in the form $ax^2 + bx + c$.

..... **[3]**

**(c)** Solve the equation $f(g(x)) = x$.
Give your answers correct to 2 decimal places.
You must show all your working.

$x =$ ..... , x =

..... **[4]**

**24** Let $f(x) = 3x + 2$, $g(x) = x^2 - x$ and $h(x) = 5^x$.

**(a)** Find $fg(x)$.
Give your answer in the form $ax^2 + bx + c$.

..... **[2]**

**(b)** Find $gf(x)$.
Give your answer in the form $ax^2 + bx + c$.

..... **[3]**

**(c)** Solve the equation

$$gf(x) = 0$$

$x =$ ..... , $x =$ ..... **[3]**

**(d)** Find the value of $x$ when $h(x) = \dfrac{1}{125}$.

$x =$ ..... **[2]**

**25** Let $f(x) = \dfrac{2x+5}{x-3}$ for $x \neq 3$, and $g(x) = 3x - 1$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Express $fg(x)$ as a single fraction in its simplest form.

$fg(x) =$ ..... **[3]**

**(c)** Solve the equation $gf(x) = x$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[5]**

**26** Consider the functions $f(x) = \dfrac{3}{x+2}$, $g(x) = (3x+5)^2$ and $h(x) = \dfrac{3x+5}{x-2}$.

**(a)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

**(b)** Find $fg(x)$.
Give your answer as a single fraction in its simplest form.

$fg(x) =$ ..... **[3]**

**(c)** Solve the equation.

$h(x) = x$

Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , or $x =$ ..... **[5]**

**27** The functions $f$ and $g$ are defined as:

$$f(x) = \frac{2x+3}{x-1} \qquad g(x) = x^2 - 4x$$

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

**(b)** Find $ff(x)$, giving your answer as a single fraction in its simplest form.

$ff(x) =$ ..... **[4]**

**(c)** Solve the equation $g(x) = f(2)$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**